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Sterilizable cover for handle of accessory of medical suspended ceiling assembly

Abstract

A sterilizable cover for a handle of an accessory of a medical suspended ceiling assembly includes a generally tubular stem portion and an annular flange portion. The generally tubular stem portion has a sufficient size to be gripped by the human hand, and has a proximal end that is open and a distal end that is closed, and defines a cavity along a longitudinal axis for receipt therein of a handle grip portion. The generally tubular stem portion includes a friction fit portion that is configured to flex radially outwardly to create a friction fit with a portion of the handle grip portion upon receipt of the handle grip portion in the cavity of the generally tubular stem portion. The annular flange portion extends radially outwardly from a proximal end of the generally tubular stem portion. The generally tubular stem portion and the annular flange portion are a single continuous piece of flexible silicone rubber.

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Background/Summary

(1) This application claims priority to U.S. Patent Application No. 63/338,070, filed May 4, 2022, which is hereby incorporated herein by reference in its entirety.

FIELD OF INVENTION

(1) This application relates generally to a cover for a handle of an accessory of a medical suspended ceiling assembly, and more particularly to a sterilizable cover for the handle of the assembly.

BACKGROUND

(2) Accessories of medical suspended ceiling assemblies typically include a handle to enable surgical personnel or other healthcare professionals to adjust the position of the accessory according to the needs of a specific medical procedure. To maintain a sterile operating environment, a sterilizable cover may be provided that fits over the handle, thus preventing the hands of surgical personnel from directly touching the handle. After the medical procedure, the sterilizable cover may then be sterilized for example in an autoclave, for later use.

(3) Current sterilizable covers have various drawbacks in certain applications. Examples of current sterilizable covers include a hard-molded outer sleeve and a two-piece sterilizable cover assembly. The hard-molded sleeve type is installed over a handle core for rigid attachment. Hard-molded outer sleeves typically require a large form factor which can result in a handle covered by such a sleeve being larger and more difficult to grasp. Moreover, where surgeon control of a light head requires capacitive touch technology, responsiveness may not be reliable with surgical glove variability and fluids commonly incurred during a medical operation. Further, hard-molded sleeves require investment in injection mold tooling, oftentimes making them cost-prohibitive in the increasingly cost-sensitive health care industry.

(4) The typical two-piece sterilizable cover may have a first piece in the form of a sterilizable metal component and a second piece in the form of a flexible insert. The sterilizable metal component includes a stem that is threaded onto a handle core and a base with openings that align with push control buttons in the handle core base. The flexible insert is sandwiched between the first piece and the handle core during assembly to cover the push control buttons and is designed to allow for interaction with the push control buttons by the flexibility in the insert. The chief shortcoming with the two-piece cover is that it requires disassembly for sterilization and then reassembly prior to being threaded back onto the handle core.

(5) Accordingly, there remains a need for further contributions in this area of technology.

SUMMARY OF INVENTION

(6) The application relates to a sterilizable cover that can be steam sterilized in a pressurized autoclave, dry heat sterilized, or chemically sterilized, after each use and reused over again. The sterilizable cover, by being made of flexible silicone rubber, is soft and flexible so that a control element of a handle underneath and covered by a corresponding portion of the sterilizable cover can be used simply by pressing the corresponding portion. A friction fit portion of the sterilizable cover creates a friction fit upon installation to retain the sterilizable cover on the handle. An annular flange portion aids in restoring the overall shape of the sterilizable cover over repeated use and sterilizations.

(7) According to one aspect of the invention, a sterilizable cover is provided for a handle of an accessory of a medical suspended ceiling assembly, the handle having a generally tubular grip portion extending downward from a mounting structure of the accessory, the sterilizable cover including a generally tubular stem portion and an annular flange portion. The generally tubular stem portion has a sufficient size to be gripped by the human hand, and has a proximal end that is open and a distal end that is closed, and defines a cavity along a longitudinal axis for receipt therein of the handle grip portion. The generally tubular stem portion includes a friction fit portion that is configured to flex radially outwardly to create a friction fit with a portion of the handle grip portion upon receipt of the handle grip portion in the cavity of the generally tubular stem portion. The annular flange portion extends radially outwardly from the proximal end of the generally tubular stem portion. The stem portion and the annular flange portion are a single continuous piece of flexible silicone rubber.

(8) Embodiments of the invention may include one or more of the following additional features separately or in combination.

(9) The generally tubular stem portion may be tapered in a direction of the longitudinal axis from the proximal end to the distal end.

(10) At least a portion of the silicone rubber may be translucent to enable the passage of light therethrough.

(11) The silicone rubber may be configured to be steam sterilized in a pressurized autoclave, dry heat sterilized, or chemically sterilized, without degradation of the silicone rubber.

(12) The silicone rubber may be configured to be resistant to temperatures between 70 and 570 degrees Fahrenheit.

(13) At least the distal end of the generally tubular stem portion, or any other portion of the sterilizable handle, may have an optically clear area configured to enable camera imaging to pass therethrough.

(14) The generally tubular stem portion may be circular in shape in axial cross section.

(15) The generally tubular stem portion may have an upper generally tubular section protruding axially downward relative to the annular flange portion and a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section.

(16) An inner perimeter of the upper generally tubular section may be relatively wider in axial cross section than an inner perimeter of the lower generally tubular section.

(17) The sterilizable may also include a shoulder that transitions radially inwardly from the inner

perimeter of the upper generally tubular section to the inner perimeter of the lower generally tubular section.

(18) An inner perimeter of the upper generally tubular section may be relatively wider in axial cross section than an outer perimeter of the lower generally tubular section.

(19) The sterilizable cover may be stackable axially with respect to another sterilizable cover such that the lower generally tubular section of one sterilizable cover is nestable along at least a portion of the longitudinal axis within the cavity of the upper generally tubular section of another sterilizable cover.

(20) The upper generally tubular section of the other sterilizable cover may be sized to receive the lower generally tubular section of the one sterilizable cover such that the bottom of the lower generally tubular section of the one sterilizable cover abuts a shoulder that transitions between the upper generally tubular section and the lower generally tubular section of the other sterilizable cover.

(21) The length of the upper generally tubular section along the longitudinal axis may be in the range of about 1 inch to about 5 inches.

(22) The length of the lower generally tubular section along the longitudinal axis may be in the range of about 1 inch to about 5 inches.

(23) The outer diameter of the upper generally tubular section may be in the range of about 1 inch to about 4 inches.

(24) The outer diameter of the lower generally tubular section may be in the range of about 0.5 inch to about 3 inches.

(25) The lower generally tubular section may include one or more channels void of material on an inner surface of the lower generally tubular section in fluid communication with the cavity of the upper generally tubular section.

(26) The friction fit portion may include at least a portion of an inner surface of the generally tubular stem portion.

(27) The generally tubular stem portion may have an upper generally tubular section protruding axially downward relative to the annular flange portion and a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, and the friction fit portion may include at least a portion of an inner surface of the lower generally tubular section.

(28) The sterilizable cover may further include at least one ridge projecting radially inward from an inner surface of the generally tubular stem portion.

(29) The generally tubular stem portion may have an upper generally tubular section protruding axially downward relative to the annular flange portion and a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, and the sterilizable cover may further include at least one ridge projecting radially inward from an inner surface of the lower generally tubular section.

(30) The at least one ridge when seen in a direction along the longitudinal axis may have a ring shape.

(31) The at least one ridge may include a plurality of ridges that when seen in a direction along the longitudinal axis form a discontinuous ring shape that is partly discontinuous.

(32) The at least one ridge may project radially inward from the inner surface to have a radially projecting width of about 0.01 inch to about 0.04 inch.

(33) The at least one ridge may have a height along the longitudinal axis of about 0.02 inch to about 0.04 inch.

(34) The at least one ridge may be located at a height along the longitudinal axis from the bottom of the generally tubular stem portion of about 1 inch to about 3 inches.

(35) A first inner surface of the generally tubular stem portion may have a first textured surface, and a second inner surface of the generally tubular stem portion may have a second textured

surface that is different from the first textured surface.

(36) The first textured surface may have a first textured finish and the second textured surface may have a second textured finish, and the first textured finish may be relatively heavier than the second textured finish.

(37) The second textured finish may be relatively smoother and less textured than the first textured finish.

(38) The second textured finish may include an optically clear finish.

(39) At least a portion of the first inner surface of the generally tubular stem portion may form part of the friction fit portion.

(40) The generally tubular stem portion may have an upper generally tubular section protruding axially downward relative to the annular flange portion and a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, and the second inner surface may include an inner surface of the upper generally tubular section.

(41) The first inner surface may include an inner surface of the lower generally tubular section.

(42) The sterilizable cover may further include at least one ridge projecting radially inward from the first inner surface of the generally tubular stem portion.

(43) The wall thickness of the generally tubular stem portion may be in the range of about 0.04 inch to about 0.16 inch.

(44) The silicone rubber of the sterilizable cover may have a Shore A hardness of about 20 to about 70.

(45) The following description and the annexed drawings set forth certain illustrative embodiments of the invention. These embodiments are indicative, however, of but a few of the various ways in which the principles of the invention may be employed. Other objects, advantages and novel features according to aspects of the invention will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The annexed drawings, which are not necessarily to scale, show various aspects of the invention.

(2) FIG. 1A is a perspective view of a sterilizable cover in accordance with an embodiment of the invention.

(3) FIG. 1B is a cross-sectional perspective view of the FIG. 1A sterilizable cover.

(4) FIG. 2A is a side elevational view of the FIG. 1A sterilizable cover.

(5) FIG. 2B is a side cross-sectional view of the FIG. 1A sterilizable cover as viewed from the plane 2B-2B in FIG. 2A.

(6) FIG. 2C is a top plan view of the FIG. 1A sterilizable cover as viewed from the plane 2C-2C in FIG. 2A.

(7) FIG. 2D is a bottom view of the FIG. 1A sterilizable cover as viewed from the plane 2D-2D in FIG. 2A.

(8) FIG. 3 is an axial cross-sectional view of the FIG. 1A sterilizable cover as viewed from the plane 3-3 in FIG. 2A.

(9) FIG. 4 is an axial cross-sectional view of the FIG. 1A sterilizable cover as viewed from the plane 4-4 in FIG. 2A.

(10) FIG. 5 is a perspective view of the FIG. 1A sterilizable cover shown installed on a handle of an accessory of a medical suspended ceiling assembly, showing light of covered illuminated control elements of the accessory handle passing through the sterilizable cover.

(11) FIG. 6 is a view showing a plurality of the FIG. 1A sterilizable covers stacked and nested.

(12) FIG. 7A is a side cross-sectional view of the FIG. 1A sterilizable cover as viewed from the plane 2B-2B in FIG. 2A, showing a friction fit of the sterilizable cover relative to a handle underneath.

(13) FIG. 7B is a side cross-sectional view of the FIG. 1A sterilizable cover as viewed from the plane 2B-2B in FIG. 2A, showing a release feature of the sterilizable cover relative to a handle underneath.

(14) FIG. 8A is a perspective view of a sterilizable cover in accordance with another embodiment of the invention.

(15) FIG. 8B is a cross-sectional perspective view of the FIG. 8A sterilizable cover.

(16) FIG. 9A is a cross-sectional perspective view of a sterilizable cover in accordance with another embodiment of the invention.

(17) FIG. 9B is a side cross-sectional view of the FIG. 9A sterilizable cover.

(18) FIG. 9C is a side cross-sectional view of the FIG. 9A sterilizable cover, showing a friction fit of the sterilizable cover relative to a handle underneath.

(19) FIG. 9D is an enlarged view of a ridge of the FIG. 9A sterilizable cover.

(20) FIG. 9E is an enlarged view of the ridge of the FIG. 9A sterilizable cover, with the sterilizable cover positioned higher relative to the handle than in the position shown in FIG. 9D.

(21) FIG. 10A is a cross-sectional perspective view of a sterilizable cover in accordance with another embodiment of the invention.

(22) FIG. 10B is a side cross-sectional view of the FIG. 10A sterilizable cover.

(23) FIG. 11A is a cross-sectional perspective view of a sterilizable cover in accordance with another embodiment of the invention.

(24) FIG. 11B is a side cross-sectional view of the FIG. 11A sterilizable cover.

(25) FIG. 12A is a cross-sectional perspective view of a sterilizable cover in accordance with another embodiment of the invention.

(26) FIG. 12B is a side cross-sectional view of the FIG. 12A sterilizable cover.

DETAILED DESCRIPTION

(27) While the present invention can take many different forms, for the purpose of promoting an understanding of the principles of the invention, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Any alterations and further modifications of the described embodiments, and any further applications of the principles of the invention as described herein, are contemplated as would normally occur to one skilled in the art to which the invention relates.

(28) FIGS. 1A-7B show a sterilizable cover **10** for a handle **20** of an accessory **22** of a medical suspended ceiling assembly in accordance with an embodiment of the invention. Herein, the sterilizable cover **10** is described in connection with a handle of a surgical light head of a suspended surgical lighting system. It is contemplated that the sterilizable cover **10** may be applicable to a handle of any type of accessory **22** of a medical suspended ceiling assembly, particularly for which an integrated handle user interface is provided for controlling the handle, the accessory, or other components of the ceiling assembly. In some embodiments, the sterilizable cover **10** may be suitable to cover a handle of a dedicated suspension arm of a medical suspended ceiling assembly.

(29) The handle **20**, which is shown with the sterilizable cover **10** installed thereon in FIGS. 5 and 7A-7B, has a generally tubular grip portion **24** extending downward from a mounting structure **26** of the accessory **22**, and an annular flange portion **28** extending radially outwardly from the grip portion **24** in proximity to the mounting structure **26**. The sterilizable cover **10** includes a generally tubular stem portion **44** and an annular flange portion **48**, which respectively cover the grip portion **24** and flange portion **28** of the handle **20**. In some embodiments, the accessory handle **20** may not include a flange portion **28**, in which case the flange portion **48** of the sterilizable cover **10** may

cover a portion of the mounting structure **26** of the accessory **22**. The generally tubular stem portion **44** of the sterilizable cover **10**, and the handle **20** on which it is installed, has a sufficient size to be gripped by the human hand meaning that the outermost diameter or perimeter of the generally tubular stem portion **44** is selected to enable a human hand to be comfortably wrapped around the generally tubular stem portion **44**. As shown in FIGS. **1A-1B**, **2A-2D** and **7A-7B**, the generally tubular stem portion **44** has a proximal end **50** that is open and a distal end **52** that is closed. The generally tubular stem portion **44** defines a cavity **60** along a longitudinal axis L-L for receipt therein of the handle grip portion **24**. The generally tubular stem portion **44** also includes a friction fit portion **70** that is configured to flex radially outwardly to create a friction fit with a corresponding portion **72** of the handle grip portion **24** upon receipt of the handle grip portion **24** in the cavity **60** of the generally tubular stem portion **44**. The annular flange portion **48** extends radially outwardly from the proximal end **50** of the generally tubular stem portion **44**. The generally tubular stem portion **44** and the annular flange portion **48** are a single continuous piece of flexible silicone rubber.

(30) As will be described in greater detail below, several advantages may be realized by the sterilizable cover **10** in accordance with the invention. For example, because the sterilizable cover **10** is constructed of silicone rubber, the sterilizable cover **10** can be steam sterilized in a pressurized autoclave, dry heat sterilized, or chemically sterilized, after each use and reused over again. Moreover, the sterilizable cover **10**, by being made of flexible silicone rubber, is soft and flexible so that a control element **80** of the handle **20** underneath and covered by a corresponding portion **82** of the sterilizable cover **10** can be used simply by pressing the corresponding portion **82**. The friction fit portion **70** creates a friction fit upon installation to retain the sterilizable cover **10** on the handle **20**. The annular flange portion **48** at the proximal end **50** of the generally tubular stem portion **44** aids in restoring the shape of the generally tubular stem portion **44** and thus the overall shape of the sterilizable cover **10** over repeated use and sterilizations.

(31) Turning initially then to FIGS. **1A-1B**, **5** and **7A-7B**, the sterilizable cover **10** is configured to be installed over an accessory handle **20** that includes, for example, integrated control elements **80**. The control elements **80** may be configured to provide an intuitive and effective user interface for the handle **20** for controlling any aspect of the system, such as attributes of the accessory for example attributes of the light emitted from the accessory-surgical light head **22**, or attributes of a camera mounted within the handle **20**, or attributes of other components of the medical suspended ceiling assembly. The control elements **80** may be in the form of push control buttons that include one or more of tactile, auditory, haptic, and visible feedback capabilities or modes. Visible feedback may be in the form of illuminated control buttons that become more bright or less bright in response to user commands. FIG. **5** shows an example of illuminated control buttons **80**.

(32) The generally tubular stem portion **44** and the annular flange portion **48** of the sterilizable cover **10** form a single continuous piece of silicone rubber that is soft to the touch. For example, the silicone rubber may have a Shore A hardness of about 20 to about 70. The single piece construction avoids the need for disassembly and reassembly of the sterilizable cover **10** for each medical procedure and sterilization process, as is the case with the afore mentioned two-piece sterilizable cover assemblies.

(33) The silicone rubber may be any desired color and may be configured to be non-reactive, stable, and resistant to temperatures between 70 and 570 degrees Fahrenheit. The material properties of silicone rubber enable the sterilizable cover to be steam sterilized in a pressurized autoclave, dry heat sterilized, or chemically sterilized, without degradation of the silicone rubber and the sterilizable cover **10** even through thousands of sterilization cycles. As such, the sterilizable cover **10** can be reused reliably.

(34) Referring next to FIGS. **2A-2D**, **3** and **4**, the generally tubular stem portion **44** is tapered in a direction of the longitudinal axis L-L from the proximal end **50** to the distal end **52** of the generally tubular stem portion **44**. As shown in FIGS. **3** and **4**, the illustrated stem portion **44** is circular in

shape in axial cross section, that is when seen in a direction along the longitudinal axis L-L, although it will be appreciated that the generally tubular stem portion **44** may have any generally tubular shape in axial cross section including generally square tubular in shape with curved corners, generally triangular tubular in shape with curved corners, and/or an elliptical tubular shape. The generally tubular stem portion **44** has an upper generally tubular section **90** protruding axially downward relative to the annular flange portion **48** and a lower generally tubular section **92** protruding axially downward relative to a bottom region **94** of the upper generally tubular section **90**. An inner perimeter **110** of the upper generally tubular section **90** is relatively wider in axial cross section than an inner perimeter **112** of the lower generally tubular section **92**. In some embodiments, the upper generally tubular section **90** may have an outer diameter in the range of about 1 inch to about 4 inches, a length along the longitudinal axis L-L in the range of about 1 inch to about 5 inches, and a wall thickness in the range of about 0.04 inch to about 0.16 inch. The lower generally tubular section **92** may have an outer diameter in the range of about 0.5 inch to about 3 inches, a length along the longitudinal axis L-L in the range of about 1 inch to about 5 inches, and a wall thickness in the range of about 0.04 inch to about 0.16 inch.

(35) In the illustrated embodiment, the sterilizable cover **10** is provided with a shoulder **98** that transitions axially downwardly and radially inwardly from the inner perimeter **110** of the upper generally tubular section **90** to the inner perimeter **112** of the lower generally tubular section **92**. Thus, in the illustrated generally tubular stem portion **44**, a width WUI, or diameter, of the inner perimeter **110** of the upper generally tubular section **90** narrows from the annular flange portion **48** to the shoulder **98**; a width WSI, or diameter, of an inner perimeter **114** of the shoulder **98** narrows from the upper generally tubular section **90** to the lower generally tubular section **92**; and a width WLI, or diameter, of the inner perimeter **112** of the lower generally tubular section **92** narrows from the shoulder **98** to the distal end **52** of the generally tubular stem portion **44**. In some embodiments, the shoulder **98** may be omitted and the sterilizable cover **10** may comprise a smooth or linear transition from the inner perimeter **110** of the upper generally tubular section **90** to the inner perimeter **112** of the lower generally tubular section **92**.

(36) The sterilizable cover **10** is configured to enable user access to existing integrated handle user interfaces of the handle **20** underneath the sterilizable cover **10**, aided in part by the silicone rubber construction of the sterilizable cover **10**. In FIG. 5, for example, the upper generally tubular section **90** incorporates the portion **82** corresponding to the control elements **80** of the handle **20** underneath. The upper generally tubular section **90**, owing to its silicone rubber construction and generally tubular geometry, is configured to flex radially inwardly in response to a user pressing the portion **82** and the underlying control element **80** and to return to an unflexed state in response to releasing of the control element **80** and corresponding portion **82**, as shown respectively in FIGS. 7A and 7B. In this way, the sterilizable cover **10** can maintain the utility of the control elements **80** of the handle **20**, for example, the afore mentioned push buttons equipped with tactile, auditory, haptic, and/or visible feedback capabilities.

(37) As noted previously, the annular flange portion **48** may aid in restoring the shape of the sterilizable cover **10**, which restoration may include restoring the upper generally tubular section **90** and the portion **82** thereof to an unflexed state. For depressible type control elements, restoring the shape of the sterilizable cover **10** may also facilitate the return of the depressible control element to its original position after release.

(38) The sterilizable cover **10** may be configured in whole or in part with silicone rubber that is translucent. For example, in FIG. 5 the upper generally tubular section **90** may be translucent. In some embodiments, the translucent portion is configured to allow a user to see the sterile control elements **80** covered by the sterilizable cover **10**. In some embodiments, the translucent portion is configured to allow the passage of light from covered illuminated components such as the illuminated control elements **80** shown in FIG. 5. In still other embodiments, and as will be described in greater detail with respect to FIGS. 8A-8B, the generally tubular stem portion **44** may

have an optically clear area **120**, or the entire stem portion **44** itself or entire sterilizable cover **10** itself may be optically clear, to enable camera imaging to pass therethrough.

(39) The sterilizable cover **10** may also be configured to be nestable and/or stackable. As shown in FIG. 2B, for example, the width WUI of the inner perimeter **110** of the upper generally tubular section **90** is relatively wider in axial cross section than a width WLO, or diameter, of an outer perimeter **122** of the lower generally tubular section **92**. In some embodiments, at least a portion of the lower generally tubular section **92** of one sterilizable cover **10** is configured to be inserted into the cavity **60** of the upper generally tubular section **90** of another sterilizable cover **10**, which may facilitate nesting and stacking of multiple sterilizable covers **10**. Referring to FIG. 6, for example, a first sterilizable cover **10A** may be stacked axially with respect to a second sterilizable cover **10B** such that the lower generally tubular section **92** of the second sterilizable cover **10B** is nested along a portion of the longitudinal axis L-L within the cavity **60** of the upper generally tubular section **90** of the first sterilizable cover **10A**.

(40) In some embodiments, the cavity **60** of the upper generally tubular section **90** of the first sterilizable cover **10A** may be sized to receive the lower generally tubular section **92** of the second sterilizable cover **10B** such that the bottom of the lower generally tubular section **92** of the second sterilizable cover **10B** abuts the shoulder **98** that transitions between the upper generally tubular section **90** and the lower generally tubular section **92** of the first sterilizable cover **10A**. FIG. 6 shows five sterilizable covers **10A**, **10B**, **10C**, **10D**, **10E** stacked and nested together in this manner. Referring to the bottom of FIG. 6, the abutting contact with the shoulder **98** also provides a gap G between adjacent stacked annular flange portions **48** of the respective sterilizable covers **10A**, **10B**, **10C**, **10D**, **10E**, enabling a user to easily position the hand between the annular flange portions **48** to separate adjacent stacked sterilizable covers **10A**, **10B**, **10C**, **10D**, **10E**.

(41) FIGS. 7A-7B show a release feature of the sterilizable cover **10** in accordance with an embodiment of the invention. As earlier described, and as shown for example in FIG. 7A, the generally tubular stem portion **44** of the sterilizable cover **10** includes a friction fit portion **70** that frictionally engages a corresponding portion **72** of the handle grip portion **24** to retain the sterilizable cover **10** on the handle **20**. In the illustrated embodiment, the friction fit portion **70** includes at least a portion of an inner surface **130** of the generally tubular stem portion **44**, and more particularly at least a portion of an inner surface **132** of the lower generally tubular section **92** of the generally tubular stem portion **44**. The release feature is configured in the following way, it being understood that other ways are also contemplated.

(42) The release feature may be incorporated into the generally tubular stem portion **44** by the relationship between the upper generally tubular section **90**, the lower generally tubular section **92**, and the shoulder **98** therebetween. As shown in FIG. 7B, the generally tubular stem portion **44** is configured such that a squeeze force SF, and in some cases also a twist force TF about the longitudinal axis L-L, applied to the upper generally tubular section **90** causes the upper part of the shoulder **98** to flex radially inwardly and, via force transfer through the shoulder **98**, imparts a radially outward force R to the lower generally tubular section **92** including at the friction fit portion **70** of the generally tubular stem portion **44**. As a result, a gap H is formed between portions of the lower generally tubular section **92** and the handle grip portion **24**, thereby to loosen the friction fit therebetween, thus releasing the sterilizable cover **10** from frictional engagement with the handle **20**, or at least reducing the frictional engagement between the inner surface **130** of the generally tubular stem portion **44** and the handle grip portion **24**, and enabling easy removal of the sterilizable cover **10** from the handle **20** by simply pulling the sterilizable cover axially downward. Once removed, the annular flange portion **48** at the proximal end **50** of the generally tubular stem portion **44** may aid in restoring the generally tubular shape of the generally tubular stem portion **44**, including the friction fit portion **70** flexing radially inwardly to its unflexed state, thereby restoring the overall shape of the sterilizable cover **10**. As will be appreciated, the upper generally tubular section **90** is configured to flex inward without “grabbing” or frictionally engaging the handle **20**

whereas grabbing the lower generally tubular section **92** reinforces the grip.

(43) FIGS. **8A-8B** show a sterilizable cover **210** in accordance with another embodiment of the invention. The FIG. **8A** sterilizable cover **210** is in many respects similar to the above-described FIG. **1A** sterilizable cover **10**, and consequently the same reference numerals are used to denote structures corresponding to similar structures in the sterilizable cover **10**. In addition, the foregoing description of the FIG. **1A** sterilizable cover **10** is equally applicable to the FIG. **8A** sterilizable cover **210** except as noted below. Moreover, it will be appreciated upon reading and understanding the specification that aspects of the sterilizable covers **10**, **210** may be substituted for one another or used in conjunction with one another where applicable.

(44) Turning then to FIGS. **8A-8B**, the sterilizable cover **210** is made of silicone rubber and is configured to cover an accessory handle that incorporates a camera therein, also referred to as a camera handle. The generally tubular stem portion **44** of the sterilizable cover **210** has an optically clear area **120** at its distal end **52** that is configured to enable camera imaging to pass therethrough. The optically clear area **120** may be circular in shape in axial cross section, as shown, or any shape suited for a camera supported within the handle covered by the sterilizable cover **210**. In the illustrated embodiment, the sterilizable cover **210** is a homogeneous design wherein the optically clear area **120**, the upper and lower generally tubular sections **90**, **92**, and the annular flange portion **48**, are made of the same silicone rubber material, that is as a monolithic one-piece unit, while the optically clear area **120** is configured to be optically clear. In some embodiments, the entire stem portion **44**, or the entire sterilizable cover **210**, may be made to be optically clear to enable camera imaging to pass therethrough. In some embodiments, surface finishing may be used to form the optically clear area **120**, or areas.

(45) The sterilizable cover **210** may also be equipped with channels **214** void of material on the inner surface **130** of the lower generally tubular section **92** to be in fluid communication with the cavity **60** of the upper generally tubular section **90**. In the FIG. **8A** embodiment, the channels **214** extend axially substantially in the direction of the longitudinal axis L-L although it will be appreciated that the channels **214** may extend along any path, linear or nonlinear, from the lower generally tubular section **92** to being in fluid communication with the cavity **60** of the upper generally tubular section **90**. The illustrated embodiment has four channels **214** equally angularly spaced apart around the circumference of the lower generally tubular section **92**. The illustrated channels **214** are 0.063 inch in width although other widths are contemplated, for example, in the range of 0.016 to 0.25 inch. In some embodiments, the channels **214** may be configured as vents. In this way, upon squeezing the upper generally tubular section **90** air may be directed from the upper generally tubular section **90** to the lower generally tubular section **92** via the channels **214** to release or reduce the frictional engagement between the sterilizable cover **210** and the handle, thereby aiding in removal of the sterilizable cover **210** from the handle. Channels **214** may be provided on the inner surface **134** of the upper generally tubular section **90** and/or the inner surface **132** of the lower generally tubular section **92** and/or an inner surface of the shoulder **98** for such venting purposes to aid in removal of the sterilizable cover **210** from the handle **20**. The afore described sterilizable cover **10** may also be provided with such channels **214**.

(46) In some embodiments, ribs may be provided on the outer or inner surface of the upper generally tubular section **90** and/or the lower generally tubular section **92** and/or the shoulder **98** to add stiffness to the sterilizable covers **10**, **210**, particularly in the direction of the longitudinal axis L-L, to aid in preventing the sterilizable covers **10**, **210** from buckling axially during installation and removal.

(47) FIGS. **9A-9E** show a sterilizable cover **310** in accordance with another embodiment of the invention. The FIG. **9A** sterilizable cover **310** is in many respects similar to the above-described FIG. **1A** and FIG. **8A** sterilizable covers **10**, **210** and consequently the same reference numerals are used to denote structures corresponding to similar structures in the sterilizable cover **310**. In addition, the foregoing description of the FIG. **1A** and FIG. **8A** sterilizable covers **10**, **210** is

equally applicable to the FIG. 9A sterilizable cover **310** except as noted below. Moreover, it will be appreciated upon reading and understanding the specification that aspects of the sterilizable covers **10, 210, 310** may be substituted for one another or used in conjunction with one another where applicable.

(48) Turning then to FIGS. 9A-9E, the sterilizable cover **310** includes at least one ridge **340** projecting radially inward from the inner surface **130** of the generally tubular stem portion **44**. In the illustrated embodiment, the ridge **340** projects radially inward from the inner surface **132** of the lower generally tubular section **92**. As will be appreciated, the ridge **340** alternately may be provided to project from the inner surface **134** of the upper generally tubular section **90**. Referring to FIG. 9A, which shows only half of the ridge **340** owing to the view being cross-sectional, the illustrated ridge **340** is ring shape in axial cross section, that is when seen in a direction along the longitudinal axis L-L. Ring shape as used herein means an annular shape, a hollow circle shape, a bracelet shape, and/or a donut shape. It will be appreciated that the ridge **340** may have any ring shape in axial cross section including square ring shape with curved corners, triangular ring shape with curved corners, and/or an elliptical ring shape.

(49) In some embodiments, the ridge **340** may project radially inward from the inner surface **130** of the generally tubular stem portion **44** to have a radially projecting width of about 0.01 inch to about 0.04 inch. Also, in some embodiments, the ridge **340** may have a height along the longitudinal axis L-L of about 0.02 inch to about 0.04 inch. The ridge **340** may be located at a height along the longitudinal axis L-L from the bottom of the generally tubular stem portion **44** of about 1 inch to about 3 inches. This enables a user to grip the lower portion of the generally tubular stem portion **44** between the thumb and index finger below the location of the ridge **340**.

(50) As shown in FIGS. 5 and 9C-9E, the generally tubular grip portion **24** may include recesses **342** that are recessed relative to an outer surface **346** of the generally tubular grip portion **24**. The illustrated embodiment has four such recesses **342** and four intermediate outer surface portions **354** equally angularly spaced apart around the outer surface **346** of the handle **20**. As shown in FIGS. 9D and 9E, the ridge **340** may be configured on the inner surface **130** of the generally tubular stem portion **44** of the sterilizable cover **310** such that as a user installs the sterilizable cover **310** onto the handle **20**, that is along the longitudinal axis L-L, the ridge **340** initially flexes radially outwardly as shown in FIG. 9D, and then as the ridge **340** is further urged upward onto the handle **20** and beyond the lower edges of the angularly spaced four recesses **342** the ridge **340** at the locations of the angularly spaced four recesses **342** unflexes at least partially, that is flexes back radially inwardly as shown in FIG. 9E. The inventors found that this radially outward and radially inward flexure of the ridge **340** provides feedback to the user that the sterilizable cover **310** is installed on the handle **20** and thus is ready for use.

(51) Like the afore described friction fit portion **70**, the ridge **340** may be configured to flex radially outwardly to create a friction fit with a portion of the generally tubular grip portion **24** of the handle **20** upon receipt of the generally tubular grip portion **24** in the cavity **60** of the generally tubular stem portion **44**. Thus, for example, in the illustrated embodiment when the ridge **340** unflexes the ridge **340** engages the four intermediate outer surface portions **354** to create a friction fit with the corresponding portion **72** of the handle grip portion **24** upon receipt of the handle grip portion **24** in the cavity **60** of the generally tubular stem portion **44**.

(52) In some embodiments, the ridge **340** may provide feedback to a user without creation of a friction fit with a corresponding portion of the generally tubular grip portion **24** of the handle **20**. For example, in an alternate embodiment, the handle may have a circumferential recess similar in radial depth to the four recesses **342** such that as the ridge **340** is urged beyond the lower edge of the recess the ridge **340** unflexes, that is flexes back radially inwardly to reside within the recess without frictionally engaging the generally tubular grip portion **24** of the handle **20**. The radially outward and radially inward flexure of the ridge **340** provides feedback to the user that the sterilizable cover **310** is installed on the handle **20** and thus is ready for use. In some embodiments,

the ridge may provide feedback to a user without recesses **342** or a recess. For example, the generally tubular grip portion **24** of the handle **20** may not include recesses **342** and the ridge **340** still provides feedback to user by the user pushing up against the sterilizable cover **310** against the ridge **340** on the inner surface **130** of the generally tubular stem portion **44**.

(53) As was noted above, the at least one ridge **340** may be provided at any portion of the generally tubular stem portion **44** whether the lower generally tubular section **92** thereof as shown or the upper generally tubular section **90**.

(54) FIGS. **10A-10B** show a sterilizable cover **410** in accordance with another embodiment of the invention. The FIG. **10A** sterilizable cover **410** is in many respects similar to the above-described FIG. **1A**, FIG. **8A**, FIG. **9A** sterilizable covers **10**, **210**, **310** and consequently the same reference numerals are used to denote structures corresponding to similar structures in the sterilizable cover **410**. In addition, the foregoing description of the FIG. **1A**, FIG. **8A**, FIG. **9A** sterilizable covers **10**, **210**, **310** is equally applicable to the FIG. **10A** sterilizable cover **410** except as noted below.

Moreover, it will be appreciated upon reading and understanding the specification that aspects of the sterilizable covers **10**, **210**, **310**, **410** may be substituted for one another or used in conjunction with one another where applicable.

(55) Turning then to FIGS. **10A-10B**, the sterilizable cover **410** includes a plurality of ridges **440** projecting radially inward from the inner surface **130** of the generally tubular stem portion **44**. In the illustrated embodiment, the plurality of ridges **440** project radially inward from the inner surface **132** of the lower generally tubular section **92**. As will be appreciated, the plurality of ridges **440** alternately may be provided to project from the inner surface **134** of the upper generally tubular section **90**. Referring to FIG. **10A**, which shows only half of the plurality of ridges **440** owing to the view being cross-sectional, the illustrated plurality of ridges **440** form in axial cross section, that is when seen in a direction along the longitudinal axis L-L, a discontinuous ring shape that is partly discontinuous. Discontinuous ring shape as used herein means a discontinuous annular shape, a discontinuous hollow circle shape, a discontinuous bracelet shape, and/or a discontinuous donut shape. It will be appreciated that the plurality of ridges **440** may have any discontinuous ring shape in axial cross section including discontinuous square ring shape with curved corners, discontinuous triangular ring shape with curved corners, and/or discontinuous elliptical ring shape.

(56) In some embodiments, the plurality of ridges **440** may project radially inward from the inner surface **130** of the generally tubular stem portion **44** to have a radially projecting width of about 0.01 inch to about 0.04 inch. Also, in some embodiments, the plurality of ridges **440** may have a height along the longitudinal axis L-L of about 0.02 inch to about 0.04 inch. The plurality of ridges **440** may be located at a height along the longitudinal axis L-L from the bottom of the generally tubular stem portion **44** of about 1 inch to about 3 inches. This enables a user to grip the lower portion of the generally tubular stem portion **44** between the thumb and index finger below the location of the plurality of ridges **440**.

(57) The plurality of ridges **440** may be configured to provide feedback to a user in the same manner as described above with respect to the ridge **340** of the FIG. **9A** sterilizable cover **310**. The plurality of ridges **440** may be configured to provide a friction fit with a portion of the generally tubular grip portion **24** of the handle **20** in the same manner as described above with respect to the ridge **340** of the FIG. **9A** sterilizable cover **310**.

(58) Like the ridge **340**, the plurality of ridges **440** may be provided at any portion of the generally tubular stem portion **44** whether the lower generally tubular section **92** thereof as shown or the upper generally tubular section **90**.

(59) FIGS. **11A-11B** show a sterilizable cover **510** in accordance with another embodiment of the invention. The FIG. **11A** sterilizable cover **510** is in many respects similar to the above-described FIG. **1A**, FIG. **8A**, FIG. **9A**, FIG. **10A** sterilizable covers **10**, **210**, **310**, **410** and consequently the same reference numerals are used to denote structures corresponding to similar structures in the

sterilizable cover **510**. In addition, the foregoing description of the FIG. **1A**, FIG. **8A**, FIG. **9A**, FIG. **10A** sterilizable covers **10**, **210**, **310**, **410** is equally applicable to the FIG. **11A** sterilizable cover **510** except as noted below. Moreover, it will be appreciated upon reading and understanding the specification that aspects of the sterilizable covers **10**, **210**, **310**, **410**, **510** may be substituted for one another or used in conjunction with one another where applicable.

(60) Turning then to FIGS. **11A-11B**, the generally tubular stem portion **44** of the sterilizable cover **510** includes a first inner surface **556** having a first textured surface **566** and a second inner surface **558** having a second textured surface **568** that is different from the first textured surface **566**. The first and second textured surfaces **566**, **568** may be formed by a texturing process, also known in the art as graining or engraving, wherein a pattern, that is a texture or grain, is added to the molding surface of a mold, which allows the mold to impress such pattern on the inner surface of the molded sterilizable cover **510**. In the FIGS. **11A-11B** embodiment, the first textured surface **566** has a first textured finish **576** and the second textured surface **568** has a second textured finish **578**. The first textured finish **576** is relatively heavier than the second textured finish **578**. In one form, the first textured finish may include a Mold Tech finish from a library of industry standard finishes for heavily textured finishes, for example, MT-11050 having a texture depth of 0.0045 inch and a draft angle of 6.5 degrees. The second textured finish **578** is relatively smoother and less textured than the first textured finish **576**. In one form, the second textured finish **578** may include an SPI (Society of Plastics Industry) standard surface finish, for example, SPI B-1 having a surface roughness (roughness average RA) of 0.05 to 0.10 mm. The second textured finish **578** may include an optically clear finish, for example, at the distal end **52** of the generally tubular stem portion **44** to aid the afore described optically clear area **120** to pass camera imaging therethrough.

(61) As shown in FIGS. **11A-11B**, the first inner surface **556** may include at least a portion of an inner surface of the lower generally tubular section **92**, for example, extending longitudinally from the top of the lower generally tubular section **92** linearly downward to just above the bend radius leading into the distal end **52** of the generally tubular stem portion **44**. Further, as shown, at least a portion of the first inner surface **556** of the generally tubular stem portion **44** may form part of the friction fit portion **70**. The second inner surface **558** may include an inner surface of the upper generally tubular section **90**. Alternately, in some embodiments, the second inner surface **558** may include all inner surfaces of the sterilizable cover **10** other than the first inner surface **556**.

(62) The first and second textured surfaces **566**, **568** of the sterilizable cover **510** enable the respective different first and second inner surfaces **556**, **558** of the sterilizable cover **510** to provide their own distinct function. For example, the first textured surface **566** aids the friction fit between the friction fit portion **70** of the lower generally tubular section **92** and the handle grip portion **24** of the handle **20** upon receipt of the handle grip portion **24** in the cavity **60** of the generally tubular stem portion **44**. The second textured surface **568**, for example, aids in allowing a user to see the sterile control elements **80** covered by the sterilizable cover **510**, see FIG. **5**, and/or to enable better the passage of light though the second textured surface **568** from any covered illuminated components of the handle **20**. The second textured surface **568**, for example, as described above, may also aid the afore described optically clear area **120** to pass camera imaging through the distal end **52** of the generally tubular stem portion **44**.

(63) FIGS. **12A-12B** show a sterilizable cover **610** in accordance with another embodiment of the invention. The FIG. **12A** sterilizable cover **610** is in many respects similar to the above-described FIG. **1A**, FIG. **8A**, FIG. **9A**, FIG. **10A**, FIG. **11A** sterilizable covers **10**, **210**, **310**, **410**, **510** and consequently the same reference numerals are used to denote structures corresponding to similar structures in the sterilizable cover **610**. In addition, the foregoing description of the FIG. **1A**, FIG. **8A**, FIG. **9A**, FIG. **10A**, FIG. **11A** sterilizable covers **10**, **210**, **310**, **410**, **510** is equally applicable to the FIG. **12A** sterilizable cover **610** except as noted below. Moreover, it will be appreciated upon reading and understanding the specification that aspects of the sterilizable covers **10**, **210**, **310**, **410**, **510**, **610** may be substituted for one another or used in conjunction with one another where

applicable.

(64) Turning then to FIGS. **12A-12B**, the sterilizable cover **610** includes at least one ridge **340** projecting radially inward from the first inner surface **556** of the generally tubular stem portion **44**. The ridge **340** is provided with the first textured surface **566** and the respective first textured finish **576**. Alternately, the FIG. **12A** sterilizable cover **610** may include a plurality of ridges **440** projecting radially inward from the first inner surface **556** of the generally tubular stem portion **44**. The plurality of ridges **440** are provided with the first textured surface **566** and the respective first textured finish **576**.

(65) As will be appreciated from the foregoing description, the sterilizable covers **10, 210, 310, 410, 510, 610** being made of silicone rubber, have a flexible form factor and are highly customizable. Thus, for example, the sterilizable covers **10, 210, 310, 410, 510, 610** can be installed directly over existing integrated handle user interfaces for example that include control elements **80**, as shown in FIGS. **5, 7A-7B**, and **9C**. Alternately, the sterilizable covers **10, 210, 310, 410, 510, 610** may be configured to be installed directly over integrated user interfaces of handles that incorporate a camera, for example as shown in FIGS. **8A-8B**, where the integrated user interfaces may control attributes of the camera itself or other components of the system.

(66) Although the invention has been shown and described with respect to a certain embodiment or embodiments, it is obvious that equivalent alterations and modifications will occur to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In particular regard to the various functions performed by the above described elements (components, assemblies, devices, compositions, etc.), the terms (including a reference to a “means”) used to describe such elements are intended to correspond, unless otherwise indicated, to any element which performs the specified function of the described element (i.e., that is functionally equivalent), even though not structurally equivalent to the disclosed structure which performs the function in the herein illustrated exemplary embodiment or embodiments of the invention. In addition, while a particular feature of the invention may have been described above with respect to only one or more of several illustrated embodiments, such feature may be combined with one or more other features of the other embodiments, as may be desired and advantageous for any given or particular application.

Claims

1. A sterilizable cover for a handle of an accessory of a medical suspended ceiling assembly, the handle having a generally tubular grip portion extending downward from a mounting structure of the accessory, the sterilizable cover comprising: a generally tubular stem portion having a sufficient size to be gripped by the human hand, wherein the stem portion has a proximal end that is open and a distal end that is closed, and defines a cavity along a longitudinal axis for receipt therein of the handle grip portion; wherein the generally tubular stem portion includes a friction fit portion that is configured to flex radially outwardly to create a friction fit with a portion of the handle grip portion upon receipt of the handle grip portion in the cavity of the generally tubular stem portion; and, an annular flange portion extending radially outwardly from the proximal end of the generally tubular stem portion; wherein the stem portion and the annular flange portion are a single continuous piece of flexible silicone rubber wherein the generally tubular stem portion has an upper generally tubular section protruding axially downward relative to the annular flange portion, a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, and a shoulder that transitions from the upper generally tubular section to the lower generally tubular section, wherein an inner perimeter of the upper generally tubular section is relatively wider in axial cross section than an inner perimeter of the lower generally tubular section, and wherein the upper generally tubular section, lower generally tubular section, and the shoulder are configured such that at least one of a squeeze force or a twist force about a

longitudinal axis of the generally tubular stem portion applied to the upper generally tubular section causes an upper part of the shoulder to flex radially inwardly and, via force transfer through the shoulder, imparts a radially outward force to the lower generally tubular section including at the friction fit portion of the generally tubular stem portion to form a gap between the lower generally tubular section and the handle grip portion, thereby to loosen the friction fit therebetween, releasing the sterilizable cover from frictional engagement with the handle, enabling easy removal of the sterilizable cover from the handle.

2. The sterilizable cover of claim 1, wherein the generally tubular stem portion is tapered in a direction of the longitudinal axis from the proximal end to the distal end.
3. The sterilizable cover of claim 1, wherein at least a portion of the silicone rubber is translucent to enable the passage of light therethrough.
4. The sterilizable cover of claim 1, wherein the silicone rubber is configured to be steam sterilized in a pressurized autoclave, dry heat sterilized, or chemically sterilized, without degradation of the silicone rubber.
5. The sterilizable cover of claim 1, wherein the silicone rubber is configured to be resistant to temperatures between 70 and 570 degrees Fahrenheit.
6. The sterilizable cover of claim 1, wherein at least the distal end of the generally tubular stem portion has an optically clear area configured to enable camera imaging to pass therethrough.
7. The sterilizable cover of claim 1, wherein the generally tubular stem portion is circular in shape in axial cross section.
8. The sterilizable cover of claim 1, wherein the shoulder transitions radially inwardly from the inner perimeter of the upper generally tubular section to the inner perimeter of the lower generally tubular section.
9. The sterilizable cover of claim 1, wherein an inner perimeter of the upper generally tubular section is relatively wider in axial cross section than an outer perimeter of the lower generally tubular section.
10. The sterilizable cover of claim 1, wherein the sterilizable cover is stackable axially with respect to another sterilizable cover such that the lower generally tubular section of one sterilizable cover is nestable along at least a portion of the longitudinal axis within the cavity of the upper generally tubular section of another sterilizable cover.
11. The sterilizable cover of claim 10, wherein the upper generally tubular section of the other sterilizable cover is sized to receive the lower generally tubular section of the one sterilizable cover such that the bottom of the lower generally tubular section of the one sterilizable cover abuts a shoulder that transitions between the upper generally tubular section and the lower generally tubular section of the other sterilizable cover.
12. The sterilizable cover of claim 1, wherein the length of the upper generally tubular section along the longitudinal axis is in the range of 1 inch to 5 inches.
13. The sterilizable cover of claim 1, wherein the length of the lower generally tubular section along the longitudinal axis is in the range of 1 inch to 5 inches.
14. The sterilizable cover of claim 1, wherein the outer diameter of the upper generally tubular section is in the range of 1 inch to inches.
15. The sterilizable cover of claim 1, wherein the outer diameter of the lower generally tubular section is in the range of 0.5 inch to 3 inches.
16. The sterilizable cover of claim 1, wherein the lower generally tubular section includes one or more channels void of material on an inner surface of the lower generally tubular section in fluid communication with the cavity of the upper generally tubular section.
17. The sterilizable cover of claim 1, wherein the friction fit portion includes at least a portion of an inner surface of the generally tubular stem portion.
18. The sterilizable cover of claim 17, wherein the generally tubular stem portion has an upper generally tubular section protruding axially downward relative to the annular flange portion and a

lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, and the friction fit portion includes at least a portion of an inner surface of the lower generally tubular section.

19. The sterilizable cover of claim 1, comprising at least one ridge projecting radially inward from an inner surface of the generally tubular stem portion.

20. The sterilizable cover of claim 19, wherein the generally tubular stem portion has an upper generally tubular section protruding axially downward relative to the annular flange portion and a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, and further comprising at least one ridge projecting radially inward from an inner surface of the lower generally tubular section.

21. The sterilizable cover of claim 19, wherein the at least one ridge when seen in a direction along the longitudinal axis has a ring shape.

22. The sterilizable cover of claim 19, wherein the at least one ridge includes a plurality of ridges that when seen in a direction along the longitudinal axis form a discontinuous ring shape that is partly discontinuous.

23. The sterilizable cover of claim 19, wherein the at least one ridge projects radially inward from the inner surface to have a radially projecting width of about 0.01 inch to about 0.04 inch.

24. The sterilizable cover of claim 19, wherein the at least one ridge has a height along the longitudinal axis of about 0.02 inch to about 0.04 inch.

25. The sterilizable cover of claim 19, wherein the at least one ridge is located at a height along the longitudinal axis from the bottom of the generally tubular stem portion of about 1 inch to about 3 inches.

26. The sterilizable cover of claim 1, wherein a first inner surface of the generally tubular stem portion has a first textured surface, and a second inner surface of the generally tubular stem portion has a second textured surface that is different from the first textured surface.

27. The sterilizable cover of claim 26, wherein the first textured surface has a first textured finish and the second textured surface has a second textured finish, and the first textured finish is relatively heavier than the second textured finish.

28. The sterilizable cover of claim 27, wherein the second textured finish is relatively smoother and less textured than the first textured finish.

29. The sterilizable cover of claim 26, wherein the second textured finish includes an optically clear finish.

30. The sterilizable cover of claim 26, wherein at least a portion of the first inner surface of the generally tubular stem portion forms part of the friction fit portion.

31. The sterilizable cover of claim 26, wherein the generally tubular stem portion has an upper generally tubular section protruding axially downward relative to the annular flange portion and a lower generally tubular section protruding axially downward relative to a bottom region of the upper generally tubular section, wherein the second inner surface includes an inner surface of the upper generally tubular section.

32. The sterilizable cover of claim 31, wherein the first inner surface includes an inner surface of the lower generally tubular section.

33. The sterilizable cover of claim 26, comprising at least one ridge projecting radially inward from the first inner surface of the generally tubular stem portion.

34. The sterilizable cover of claim 1, wherein the wall thickness of the generally tubular stem portion is in the range of 0.04 inch to 0.16 inch.

35. The sterilizable cover of claim 1, wherein the silicone rubber has a Shore A hardness of 20 to 70.
